TOPIC 44/60

Document Ingestion in RAG

> The ETL of the AI Era.
>
> How we transform messy, unstructured enterprise data into clean, searchable, mathematical memory.

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THE CHALLENGE

The Unstructured Chaos

Vector Databases only understand numbers (embeddings). But human knowledge is trapped in PDFs, Confluence pages, Word docs, raw HTML, and Slack messages.

The "GIGO" Principle (Garbage In, Garbage Out)

If you feed a RAG system poorly extracted text containing broken tables, misaligned headers, or unreadable OCR, the AI will fail spectacularly—regardless of how smart the LLM is.

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THE BIG PICTURE

The 5-Step Ingestion Pipeline

1 **Parsing:** Extracting raw text from various file formats.

2 **Cleaning:** Removing boilerplate, noise, and formatting errors.

3 **Chunking:** Slicing text into contextually sound pieces.

4 **Embedding:** Converting chunks into mathematical vectors.

5 **Indexing:** Storing vectors + metadata into the database.

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STEP 1

Parsing & Extraction

The hardest technical challenge in RAG. A PDF is not a string of text; it's a set of visual drawing instructions for a printer.

Text Extractors

PyPDF, pdfplumber. Good for simple documents, but fails entirely on complex multi-column layouts or tables.

Vision & OCR

Tools like Unstructured.io or LlamaParse use AI vision models to look at the document and intelligently extract tables and charts as Markdown.

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STEP 2

Cleaning & Preprocessing

Raw extracted text is incredibly noisy. If we embed the noise, the LLM will hallucinate.

RAW EXTRACTION (NOISE)

```
CONFIDENTIAL PAGE 42
The company grew by 14% \n \n \n \n
Copyright 2026. All rights reserved.
Q3 Report
```

CLEANED DATA

```
The company grew by 14%. Q3 Report.
```

* Removed headers, footers, consecutive whitespace, and boilerplate legalese.

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STEP 3

Chunking (Recap)

We cannot embed a whole book into one vector. We slice the cleaned text into semantically relevant pieces.

A

Determine Size

Typically 256 to 1000 tokens depending on the embedding model's capacity and the complexity of the data.

B

Apply Strategy

Use Recursive Character, Markdown structural splits, or advanced Semantic boundary detection.

C

Add Overlap

Always overlap chunks by 10-20% to prevent "context shearing" (losing the meaning between cuts).

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STEP 4

Vector Embedding

This is where the magic happens. We pass every single text chunk through an Embedding Model (like OpenAI's text-embedding-3-small).

```
Input Chunk:
"To reset the password, click the blue link."

Output Vector (e.g., 1536 Dimensions):
[0.0124, -0.0531, 0.9932, -0.1102, 0.0431, -0.9991, 0.3341, -0.4412, ... 1528 more floats]
```

These coordinates mathematically represent the *concept* of resetting a password.

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STEP 5

Metadata & Indexing

Vectors alone are not enough. We must attach **Metadata** to the chunk before saving it to the Vector DB to enable *Hybrid Search*.

```
// Payload saved to DB
{
  "id": "chunk_994",
  "vector": [0.012...],
  "text": "Policy active...",
  "metadata": {
    "source": "HR_v2.pdf",
    "date": "2026-07-16",
    "department": "Sales"
  }
}
```

Why Metadata matters?

If a user asks: *"What is the new remote work policy for Sales?"*

The DB will first filter by department == "Sales" and date > 2025, THEN perform the vector search. This guarantees precision.

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THE GOLDEN RULE

The 80/20 Rule of RAG

80% of your time will be spent building the Ingestion Pipeline.

Only 20% is spent tuning the LLM.

Continuous Sync: In production, Ingestion is not a one-time script. It is an automated, event-driven pipeline (like Airflow or Dagster) that constantly monitors folders, updates vectors, and deletes obsolete chunks when original files change.

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Document Ingestion

CONCEPT UNLOCKED

👁️ **Parsing is hard; use vision models for messy PDFs**

🧹 **Clean out boilerplate to prevent hallucination**

🧩 **Chunk properly to preserve semantic boundaries**

🏷️ **Always attach metadata to unlock Hybrid Search**

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